Exercise Sheet 6: Linear Transformations

December 2024

1. Find the image of v and the preimage of w for the linear transformation:

   (a) $T\colon \mathbb{R}^2 \to \mathbb{R}^2$, $T(v_1, v_2) = (v_1, v_1 + 2v_2)$, $v = (2, -3)$, $w = (4, 12)$;

   (b) $T\colon \mathbb{R}^3 \to \mathbb{R}^3$, $T(v_1, v_2, v_3) = (0, v_1 + v_2, v_2 + v_3)$, $v = (-3, 2, 5)$, $w = (0, 2, 5)$.

2. Determine whether the function is a linear transformation. If it is, find its stanadard matrix A.

   (a) $T\colon \mathbb{R}^2 \to \mathbb{R}^2$, $T(x_1, x_2) = (x_1 + 2x_2, -x_1 - x_2)$;

   (b) $T\colon \mathbb{R}^2 \to \mathbb{R}^2$, $T(x_1, x_2) = (x_1 + 3, x_2)$;

   (c) $T\colon \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (x + y, 2y)$;

   (d) $T\colon \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (|x|, |y|)$;

   (e) $T\colon \mathbb{R}^3 \to \mathbb{R}^3$, $T(x, y, z) = (z, y, x)$.

3. Let T be a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ such that $T(2, 0) = (1, 1)$ and $T(0, 3) = (3, 3)$. Find $T(1, 1)$ and $T(1, 3)$.

   In Exercises 4-7, the linear transformation $T\colon \mathbb{R}^n \to \mathbb{R}^m$ is defined by $T(v) = Av$. For each matrix A, (a) determine the dimensions of $\mathbb{R}^n$ and $\mathbb{R}^m$, (b) find the image $T(v)$ of the given vector v, and (c) find the preimage of the given vector w.

4. $A = \begin{bmatrix} 0 & 1 & 2 \\ -2 & 0 & 0 \end{bmatrix}$, $v = (6, 1, 1)$, $w = (3, 5)$.

5. $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$, $v = (8, 4)$, $w = (5, 2)$.

6. $A = [2 \; -1]$, $v = (1, 2)$, $w = (-1)$.

7. $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \\ 1 & 2 \end{bmatrix}$, $v = (1, 2)$, $w = (2, -5, 12)$.

   In Exercises 8-11, find a basis for (a) $\ker(T)$ and (b) $\text{range}(T)$.

8. $T\colon \mathbb{R}^3 \to \mathbb{R}^3$, $T(x, y, z) = (x, y + 2z, z)$.

9. $T\colon \mathbb{R}^3 \to \mathbb{R}^3$, $T(x, y, z) = (x + 2y, y + 2z, z + 2x)$.

10. $T\colon \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (x + 2y, -2x - 4y)$.

11. $T\colon \mathbb{R}^4 \to \mathbb{R}^3,\ T(x, y, z, w) = (4x + 5y + 5z + 2w, -2x + 2y - w, 8y + 4z)$.

In Exercises 12-15, the linear trasformation T is given by $T(v) = Av$. Find a basis for (a) the kernel of T and (b) the range of T, and then find (c) the rank of T and (d) the nullity of T.

12. $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 1 & 0 \\ 0 & 1 & -3 \end{bmatrix}$.

13. $A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$.

14. $A = \begin{bmatrix} 1 & 2 \\ -1 & -2 \\ 1 & 1 \end{bmatrix}$.

15. $A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 3 & 1 & 2 & -1 \\ -4 & -3 & -1 & -3 \\ -1 & -2 & 1 & 1 \end{bmatrix}$.

16. Given $T\colon \mathbb{R}^5 \to \mathbb{R}^3$ and $\text{nullity}(T) = 2$, find $\text{rank}(T)$.

17. Given $T\colon M_{2,2} \to M_{2,2}$ and $\text{rank}(T) = 3$, find $\text{nullity}(T)$.

18. Given $T\colon P_5 \to P_3$ and $\text{nullity}(T) = 4$, find $\text{rank}(T)$.

In Exercises 19-22, determine whether the transformation T has an inverse. If it does, find A and A^{-1}.

19. $T\colon \mathbb{R}^2 \to \mathbb{R}^2,\ T(x, y) = (2x, y)$.

20. $T\colon \mathbb{R}^2 \to \mathbb{R}^2,\ T(x, y) = (x + y, 2x + 2y)$.

21. $T\colon \mathbb{R}^3 \to \mathbb{R}^2,\ T(x, y, z) = (x + y + z, z)$.

22. $T\colon \mathbb{R}^3 \to \mathbb{R}^3,\ T(x, y, z) = (x, x + y, x + y + z)$.

23. Determine whether the linear transformation T in Exercise 8 is (a) one-to-one, (b) onto and (c) invertible.

24. Determine whether the linear transformation T in Exercise 13 is (a) one-to-one, (b) onto and (c) invertible.

25. Find the standard matrices for $T = T_1 \circ T_2$ and $T' = T_2 \circ T_1$, where

$$T_1 \colon \mathbb{R}^2 \to \mathbb{R}^3, T_1(x, y) = (x, x + y, y),$$
$$T_2 \colon \mathbb{R}^3 \to \mathbb{R}^2, T_2(x, y, z) = (0, y).$$

26. Find $T(v)$ by using (a) the standard matrix and (b) the matrix relative to B and B', where

$$T \colon \mathbb{R}^2 \to \mathbb{R}^3,\ T(x, y) = (-x, y, x + y), v = (0, 1),$$
$$B = \{(1, 1), (1, -1)\}, B' = \{(0, 1, 0), (0, 0, 1), (1, 0, 0)\}.$$

27. Find the matrix A' for T relative to the basis B', where

$$T \colon \mathbb{R}^2 \to \mathbb{R}^2, T(x, y) = (x - 3y, y - x),\ \ B' = \{(1, -1), (1, 1)\}.$$